BIOT 6900 Module 4: Structure-Based Druggability

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Disease: Alzheimer's (AD)

Introduction

In this module I examined APOE (P02649) and HCN1 (O60741) for their druggability using AlphaFold confidence metrics pLDDT and PAE, similarity to an experimental structure, and the detection of pockets using fpocket. I chose APOE because it was the top-ranked protein in module 2, and though the tractability score in module 3 lowered the ranking due to the low p_sm (0.002) and p_ab (0.016), I was curious to examine if the structure would make it a viable candidate. I chose HCN1 given that it was highly ranked in both module 2 (2nd) and module 3 (1st). I compared APOE with 1GS9 chain A[1] and HCN1 with 8Y60 chain A[2]. Ultimately, I made the call that APOE was possible but uncertain, and the HCN1 structure supports a small-molecule pocket.

p53 Anchor Check

Compared to 1TUP chain B, p53 has an RMSD of 0.73 Å when for the regions over the common residues and 0.68 Å over those with pLDDT above 70. Residues 121, 183, and 187, which had the largest distance, are surface loops which contact DNA, something not shown in the model[3]. For the ligands, C176, H179, C238 and C242 are zinc ligands, while R248 and R273 contact DNA. My module 3 anchor was Y220, and it scored -3.8, which makes sense, given that it is 29 Å away from the zinc ligands and the DNA. Y220 causes damage by destabilizing the protein at a non-active site, which is not something found in the sequence score[4].

Which Parts to Trust

For this, I used the pLDDT and PAE numbers for the analysis of trustworthy domains to trust in each protein.

First, I examined APOE. The low regions are 1–29 and 31–40, with 1-18 being the signal peptides on the protein chain that are cleaved before the APOE is secreted. The rest are the beginning part of the mature chain. This meant that 1GS9 numbering was 18 off of the UniProt numbering. Therefore, I aligned the pairs correctly using this offset of 18. APOE then proceeds to domain 1 which consists of multiple helices between 42 and 208. The next domain is between 226 and 303, a lipid-binding domain. The APOE showed a mean pLDDT of 75.5, with 31.9% above 90 and 16.1% below 50. The PAE analysis showed two confident domains, which were the two I described above. The space between the domains has a mean PAE of 25.82 Å. Because this is high, pockets in this region cannot be trusted.

Then, I examined HCN1. HCN1 only had one confident region according to PAE, which was between 92-634. This area spans the HCN region, the S1-S6 transmembrane regions, the C-linker, and the CNBD[6][7]. The mean pLDDT was 68.9, with 39.0% above 90 and 37.9% below 50. The mean was certainly lowered by long tails, 1–85, 637–857 and 863–890, on either side of the confidence interval.

Given the analysis from both, the regions to trust are the confidence domains found for each protein, 42-208 and 226-303 for APOE and 92-634 for HCN1.

Matching with the Experiment

To compare to the experiment, I looked at APOE vs 1GS9 chain A. There were 144 aligned pairs with 134 of them being identical and confident. The RMSD was 0.77 Å for domain 1 (42-208), and the RMSD was 0.77 Å over the 134 identical residues. The largest deviations were residues 97-105, an area between helices in the first domain[5]. It is a surface loop, so it is likely the deviation was caused by its flexibility rather than prediction error.

Then I looked at HCN1 vs 8Y60 chain A. There were 473 aligned pairs with 459 identical and confident and an RMSD 1.35 Å. The largest deviations were in 470-477 and 204-205. 470-477 is near the C-linker,[6][7] which is packed against a neighboring unit, leading to the less accurate prediction.

Druggability Calls

I called APOE's druggability possible but uncertain. The only pocket that clears the druggability score cutoff of 0.5 was pocket 20 with a score of 0.546, with the others all having scores of 0.005 or less. Pocket 20 is kept, but the pLDDT confidence is 76.5, which is just on the border of being meaningful. However, it does have a PAE of 4.0Å, which means the residues are confidently placed in space. Unfortunately, part of the pocket (31-39) is low-confidence, which means the mean pLDDT of 76.5 is actually hiding an area of low confidence which is also not included in the experimental check.

Table 1: Top 5 Druggability

Gene	Pocket	Rank	Druggability score	Volume (Å³)	Mean pLDDT	PAE within (Å)	Verdict
APOE	20	1	0.546	911	76.5	4.0	keep
APOE	2	2	0.005	250	88.1	1.9	keep
APOE	5	3	0.005	382	73.9	4.8	keep
APOE	8	4	0.005	618	56.3	7.6	reject

APOE	3	5	0.002	237	71.9	4.1	keep
HCN1	2	1	0.878	407	92.7	2.1	keep
HCN1	7	2	0.418	488	84.7	3.2	keep
HCN1	1	3	0.245	1160	86.3	3.7	keep
HCN1	8	4	0.212	544	92.7	2.0	keep
HCN1	6	5	0.128	385	83.4	4.0	keep

I called HCN1 as a structure (which) supports a small-molecule pocket. All of the top five pockets are kept. These top five pockets also have pLDDTs ranging from 83-93 and PAEs ranging from 2.0-4.0 Å. Experimentally, the drug in 8Y60, ivabradine, is in pocket 7, which was the rank 2 pocket and covers 75% of the drug site. Therefore, there is evidence that HCN1 supports a small-molecule pocket.

Structural Call vs Multiomics

For APOE, the multi-omics analysis ranked the protein first, which is consistent with the fact that it is a strong genetic risk factor for AD. Through the prior and structural analysis, however, the ranking has gone down, and the druggability is in question. I believe that it is still worthwhile to keep this target, given that it is a well-known indicator of AD, but the evidence has been mixed.

For HCN1, the rank began at number 2 for the multi-omics analysis, and then went up to first place via the sequence prior. Given the actual evidence of a small molecule pocket and the confident structural modeling indicating the pocket, it has become a very interesting line of study to keep pursuing.

Limitations

One major limitation in the analysis of APOE was that there were missing pieces. The function of APOE depends heavily on the lipid it binds to, something that is entirely missing in this structural analysis[5]. If APOE were modeled bound to a lipid, this would improve the model.

A major limitation for the analysis of HCN1 is that there is no membrane in the model. It is a transmembrane model, and the drug site is partially covered by the phospholipid bilayer, leading to physical limitations of druggability that are not analyzed here[2]. If membrane-bound HCN1 were used, the analysis would be more accurate.

Best Pocket Analysis

For APOE, 1GS9 chain A has no small-molecule ligand, so there is nothing to check. For HCN1, 8Y60 has two ligands to check. The best pocket was pocket 2. VNZ A902, the drug ivabradine, does not lay at pocket 2, however it lays at the second best pocket, pocket 7, with 75% coverage. The other ligand, CLR A901, also does not sit at the top pocket, but at the 16th best pocket 61. The actual pocket was scored fairly low by fpocket, which could be in part given the lack of context. Without the membrane being present in the model, the pocket could look too open and unprotected to be viable.

Conclusion

HCN1 has a confident pocket for druggability backed by experimental data. APOE has one site that scores as marginally druggable but with no other evidence to back it up.

Sources

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[3] Cho Y, Gorina S, Jeffrey PD, Pavletich NP. Crystal structure of a p53 tumor suppressor–DNA complex: understanding tumorigenic mutations. Science. 1994;265(5170):346–355. doi:10.1126/science.8023157

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